Samuel Yamanaka Lownie

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Education

Queen's University Kingston, ON

Bachelor of Arts in Computer Science (Honour's) *Sep. 2020 – Apr. 2024*

Relevant Coursework: Data Structures, Algorithms, Databases, UI/UX Design

Extracurriculars: COMPSA - Scribe, Queen's Web Dev Club - Member, Queen's Game Dev Club - Member

Experience

Software Engineer (Volunteer) Dec. 2025 – Apr. 2026

Tomkulak Consortium *Canada*

- Integrated Mapbox GL into an existing React.js project and implemented pan, zoom, and viewport controls.
- Created a mock REST API using Express.js that served mock GIS data for testing purposes.
- Developed a layer management system that consumed GeoJSON from the mock API and rendered infrastructure assets as map layers.

Programming Mentor Sep. 2024 – Present

First Robotics Competition (FRC) Team 5024 *London, ON*

- Created and presented beginner-friendly tutorials that taught an overview of Java and WPILib over the span of 13 weeks.
- Maintained a 100% retention rate with students and reduced workload of other mentors by 30%.
- Guided multiple groups of students and utilized an Agile approach to ensure deadlines were met.

Customer Service Representative May 2022 – Sep. 2023

Liquor Control Board of Ontario (LCBO) *London, ON*

- Gained a deep knowledge of products sold and provided expert advice to customers within the first week.
- Prioritized and adapted to the needs of a hands-on, fast-paced environment, while remaining flexible during peek periods.
- Developed strong relationships with co-workers and provided friendly, professional service to customers.
- Implemented a new method of unloading stock which reduced task completion time by 30% within the first 2 months of role.

Projects

Active Turn Based Combat Engine | *C#, Godot*

- A turn-based system where turns are determined in real-time, based on GDQuest's OpenRPG system, and translated into C#.
- Utilized the Observer pattern for passive effects and the Composition pattern for data type creation.
- Implemented a modular effect system allowing for hundreds of unique skill combinations.

Hiyoko | *Vue.js, TypeScript, Node.js, Express.js, MongoDB, Vitest, Playwright*

- Developed a full-stack web application using JavaScript with Node.js/Express.js serving as a REST API with Vue.js as the frontend.
- Utilized a MVC design pattern with MongoDB for data retrieval and Pinia for handling complex user state.
- Reduced 200+ HTML files down to 3 using JSON and Vue.js's templating system.

Firmware Vulnerability Analyzer | *Python, PyTorch, JavaScript, React, Socket.IO, Chakra UI, Docker Compose, Linux*

- A web-interface used to detect and summarize bugs and exploits in a given firmware image.
- Utilized an agile development process, emphasizing collaboration, adaptability, and incremetnal progress.
- Devised a method to allow the project to be run on Windows and Linux machines, updated the project to use Poetry 1.7, adjusted the UI with responsive web design principles in mind, and fixed a bug related to AMD GPUs.

Skills

Languages: Java, Python, JavaScript/TypeScript, HTML/CSS, SQL, PHP
Frameworks: React, Vue.js, Express.js
Technologies: Git, Node.js, REST APIs, MySQL, MongoDB, Docker
Soft Skills: Willing to Learn, Independent, Adaptable, Strong Communicator, Team Player